

DEVELOPING-AN-ANDROID-APPLICATION-TO-IMPLEMENT-A-TOKEN-DISPENSER-SYSTEM.

AIM:

To develop an android application to perform Token dispenser

APPARATUS REQUIRED:

ØComputer ØAndroid studio software

THEORY:

The Token Dispenser App generates unique tokens for users in a queue-like system. The app allows users to input their name and receive a token number upon clicking a button. The token is displayed, and a list of previously generated tokens is maintained and shown in a ListView. This experiment demonstrates how to manage dynamic lists and use array adapters in Android for displaying data. It also introduces handling user input, generating unique tokens, and managing collections like arrays and lists. The Token Dispenser App is an excellent introduction to queue management and creating interactive applications where users can receive sequential outputs.

PROCEDURE:

1. Open Android Studio and then click on File -> New -> New project. 46. Then type the application name as “ex.no.5” and click Next.
2. Then select the Minimum SDK as shown below and click next. 48. Then select the Empty Activity and click next.
3. Finally click Finish.
4. Click on app -> java -> com.example -> MainActivity.java
5. Now click on Text and type the program, so now the programming part of the main activity is completed.
6. Click on app -> res -> layout -> activity_main.xml.
7. Now click on Text and type the program, so now the designing part of Activity main is also completed.
8. Select the suitable available device to display the output. 55. Now run the application to see the output.

PROGRAM:

MainActivity.java

```
package com.example.homeapplication;

import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    TextView fanStatus, lightStatus, acStatus;
    Button fanOn, fanOff, lightOn, lightOff, acOn, acOff;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

```

});

// Linking UI elements
fanStatus = findViewById(R.id.fanStatus);
lightStatus = findViewById(R.id.lightStatus);
acStatus = findViewById(R.id.acStatus);

fanOn = findViewById(R.id.fanOn);
fanOff = findViewById(R.id.fanOff);
lightOn = findViewById(R.id.lightOn);
lightOff = findViewById(R.id.lightOff);
acOn = findViewById(R.id.acOn);
acOff = findViewById(R.id.acOff);

// Button functionality
fanOn.setOnClickListener(v -> fanStatus.setText("Fan: ON"));
fanOff.setOnClickListener(v -> fanStatus.setText("Fan: OFF"));

lightOn.setOnClickListener(v -> lightStatus.setText("Light: ON"));
lightOff.setOnClickListener(v -> lightStatus.setText("Light: OFF"));

acOn.setOnClickListener(v -> acStatus.setText("AC: ON"));
acOff.setOnClickListener(v -> acStatus.setText("AC: OFF"));
}
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:id="@+id/titleText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Appliance Control Panel"
            android:textSize="24sp"
            android:textStyle="bold"
            android:gravity="center"
            android:paddingBottom="20dp" />

        <!-- Fan Control -->
        <TextView

```

```
        android:id="@+id/fanStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Fan: OFF"
        android:textSize="18sp"
        android:padding="8dp" />

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/fanOn"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
        android:text="Fan ON" />

    <Button
        android:id="@+id/fanOff"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
        android:text="Fan OFF" />

</LinearLayout>

<!-- Light Control -->
<TextView
    android:id="@+id/lightStatus"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Light: OFF"
    android:textSize="18sp"
    android:padding="8dp" />

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/lightOn"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
        android:text="Light ON" />

    <Button
        android:id="@+id/lightOff"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
```

```
        android:text="Light OFF" />
    </LinearLayout>

    <!-- AC Control -->
    <TextView
        android:id="@+id/acStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="AC: OFF"
        android:textSize="18sp"
        android:padding="8dp" />

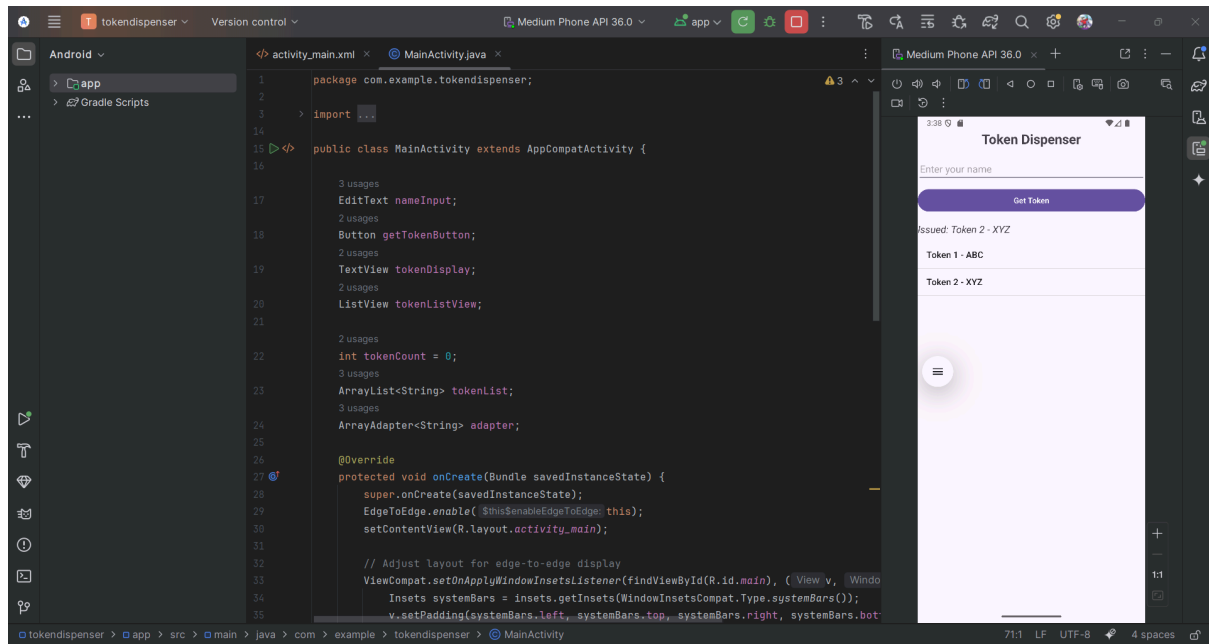
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <Button
            android:id="@+id/acOn"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:text="AC ON" />

        <Button
            android:id="@+id/acOff"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:text="AC OFF" />
    </LinearLayout>

</LinearLayout>
</ScrollView>
```

OUTPUT:



RESULT:

Thus, the Android app for generating and tracking token numbers in a queue is developed and the output is verified.